

Lab Report 3: Coherence and Interference

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1 Executive Summary

The lab explores the principles of coherence and interference using an optical setup consisting of a Michelson interferometer. Initially, the lab explores how varying different elements in the optical set up affect diffraction patterns. Particularly, controlling mirror displacement lengths, refractive index changes with a lighter, and the two output effects on patterns. The second part of the experiment explored how rotating a plexiglass plate affects fringe counts, estimating the refractive index effects by rotating the mirror in respect to the interferometer. The last part of the lab inserted red and white leds into the optical setup replacing the laser. According to theory the white led displays a tighter coherence region than the red region. The lab displays results to show the effect of optical path adjustment, differing wavelengths, and coherence affects fringe counts in interference patterns.

From the interference equation the phase difference ϕ from two different waves governs constructive and destructive interference.

In the Michelson interferometer, the optical path length of the arms affects the phase difference. A beamsplitter takes the initial source of light and separates it into two different optical arms. The split ray then reflects off mirrors and recombines at the beamsplitter. The split rays have differing E fields from phase change:

$$E_1 = \sqrt{RT} \cdot E_0 \cos(\omega t + \phi_1) \quad (1)$$

$$E_2 = \sqrt{RT} \cdot E_0 \cos(\omega t + \phi_2) \quad (2)$$

They recombine at the beamsplitter with an average intensity of:

$$I = \frac{1}{4} c \epsilon_0 E_0^2 (1 + \cos(\Delta\phi))$$

Equations 1-3 Thorlabs pg. 19, 20

2 Introduction

Interference occurs when multiple waves superpose. When two waves overlap their intensity depends on amplitude and phase.

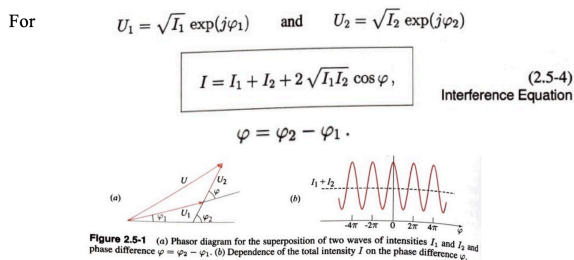


Figure 1: Interference Equation from Section 4, slide 6

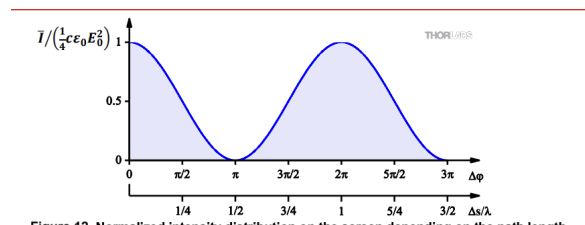


Figure 2: Normalized intensity distribution. Thorlabs pg 21

The beamsplitter introduces two different phase shifts which correspond with the two out ports on the Michelson interferometer. The ports differ by a phase of pi. Visually the bright spots

on the fringe pattern for one port corresponds to dark spots on the second port.

Visually, the circular fringe pattern appears from colliding spherical waves. When the arms are varying lengths the spherical waves interfere and produce a pattern.

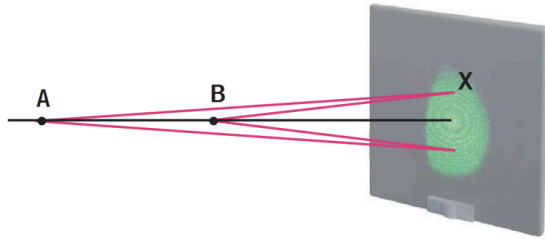


Figure 3: Circular Interference Pattern Thorlabs pg 22

Tilting the mirrors in an optical setup changes the nature of the spherical wave. The changed spherical wave causes a slight change in optical path length resulting in changes in the fringe pattern. Small changes in the optical setup involving a Michelson interferometer greatly vary the resulting fringe patterns because wavelengths are on the order of nanometers.

Once the beam hits the interferometer it travels to the mirror and back in its arm length. Therefore, changing the displacement of the mirror affects the total path displacement by a factor of 2 (there and back):

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta s \quad (4)$$

Equation 4 Thorlabs pg 20

Changing the path difference (s), affects the phase difference which produces the varying dark and light patterns of destructive and constructive interference on the visual fringes.

Each transition in the fringe pattern is a difference of λ corresponding to varying mirror displacement $\lambda/2$

$$\lambda = \frac{2\Delta x}{N} \quad (5)$$

Equation 5 Thorlabs pg 23

Interference only appears when the beam is stable and maintains its phase. The stability in the beams corresponds to coherence length which indicates the longest length where light still remains correlated. A light source with a small enough width has a coherence length of:

$$L_c \approx \frac{\lambda_0^2}{\Delta\lambda} \quad (6)$$

Equation 6 - Thorlabs pg 25

Lasers have narrow spectral widths ($\Delta\lambda$) which means their coherence length is long and they can maintain phase length. LEDs have a varying range of coherence lengths; therefore, based on the wavelength of the LED the mirror displacement at a very small scale can result in the fringe pattern disappearing.

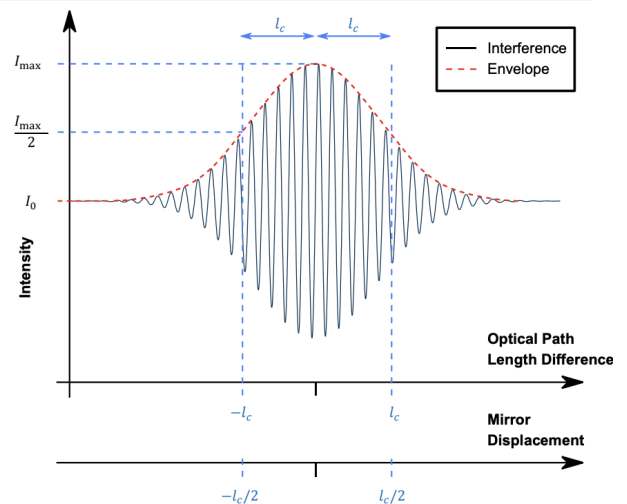


Figure 4: Envelope interference vs coherence length, Thorlabs pg 26

3 Experimental Procedure and Data

3.1 Preliminary Test and Determining Laser Wavelength

Procedure: The optical setup initially consists of a laser, Michelson interferometer, and two mirrors.

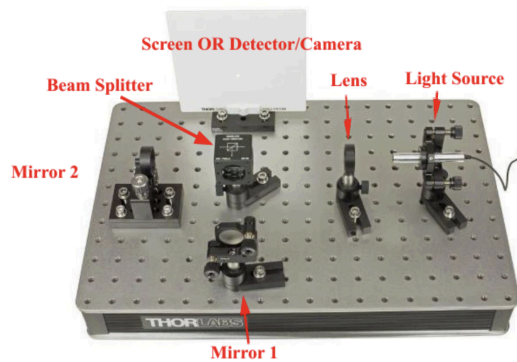
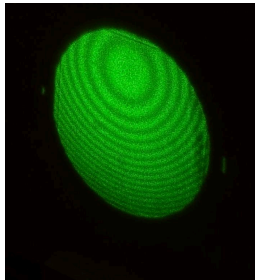


Figure 5: 3.1 setup

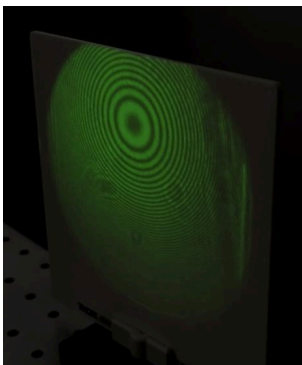
After all components are aligned by their optical axis an interference pattern should appear on the screen. First, translate one mirror and note the differences on the fringe pattern displayed on the screen as well as how it changes in visibility.



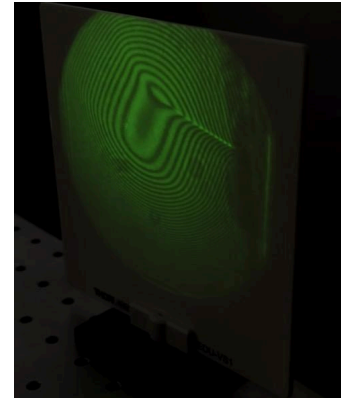
3.1 part 1

An interference pattern was captured showing destructive (dark) and constructive (light) fringes.

Next, place a lighter in one of the arms of the Michelson interferometer and notice the effect on the fringe pattern displayed on the screen.

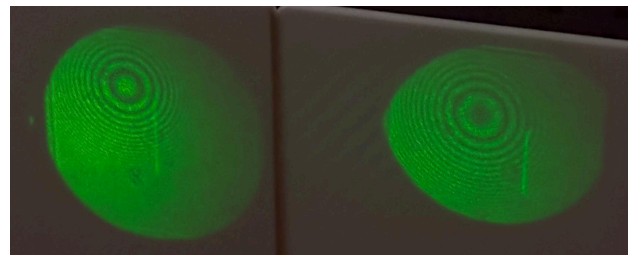


Before lighter was introduced in arm



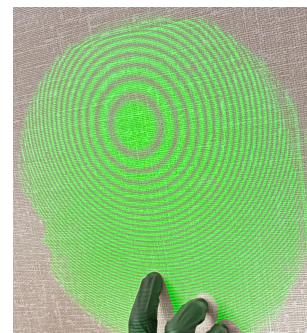
After lighter was introduced in arm

After place the screen at both ports of the Michelson interferometer and note the differences in fringe patterns.

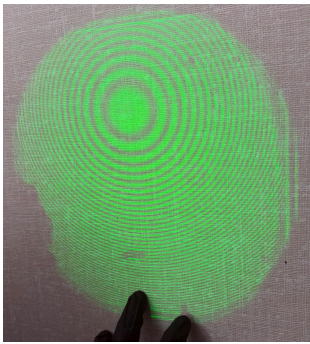


Port 1 and Port 2 fringe patterns

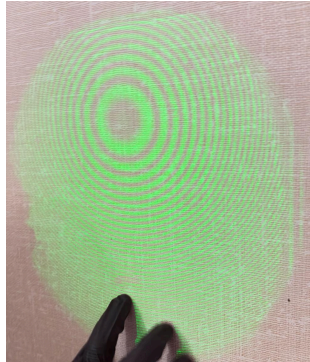
Finally, change the distance of one mirror and observe the number of fringes at fixed locations and count the number of resolvable bright fringes.



49 fringes - 0 tick displacement



45 fringes - 15 tick displacement



46 fringes - 20 tick displacement

3.2 Refractive Index Determination

Procedure: Keeping the same set up, place a rotational 8mm thick plexiglass mirror near the arm of the interferometer. First, regain the interference pattern on the screen by aligning the optical pieces properly. The plate needs to be perpendicular to the beam.

The plexiglass should be rotated in angular increments and the fringe pattern for these rotations should be noted.

There was no power meter available to measure the fringe movements.

Data collected:

0 degree rotation - 55 fringes

6 degree rotation - 75 fringes

3.3 Interference with LEDs

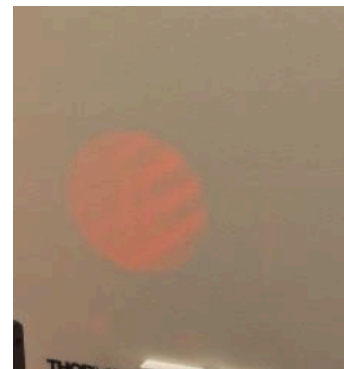
Procedure: Follow the steps from Part 1 again and re align the laser so that a fringe pattern appears on the screen. After a resolvable fringe pattern appears, turn off the laser and add a red LED to the set up acting as the light source now.

Adjust the mirror 2 with its swivel and record the full distance where the fringe pattern remains resolvable.

Next replace the red LED with a white LED and repeat the steps measuring the full range of distances where the fringe is resolvable.

Data:

Because the setup and fringe patterns depend on a matter of microns, the fringes were not captured in the optical setup. Theoretically, the fringes should appear as described in the Thorlabs manual.



Thorlabs pg 39



Thorlabs pg 41

4 Discussion, Analysis

4.1 Preliminary Test and Determining Laser Wavelength

Because the beam travels from the Michelson interferometer to the mirror and back from the mirror to the interferometer, displacing one mirror by a distance $s/2$ changes the path difference by s . According to theory, when the path difference is changed by one wavelength the fringe pattern should go through a cycle. As the mirror was displaced further it reached the edge of the coherence and the fringes started to blur out and disappear.

When the lighter was placed in a singular arm the fringe pattern noticeably contorted. By lighting an area in the arm's distance, the particles are excited and start to move faster in the region. As the beam enters the selected region it refracts and produces pulsing fringe patterns. This is because the flame is not constant and the region experienced rapid tiny changes in refraction index.

When the fringe pattern was captured at both ports it obeyed theory. The images were perfect contrasts of one another. It is extremely apparent near the central fringe bands where on the first the fringe pattern is bright and its contrasting second port is completely dark. This explains the π phase shift in destructive and constructive behavior of the fringe patterns.

Determining the wavelength proved to be difficult because the swivel on the mirror had no documentable unit to measure. Instead the fringe patterns were counted at fixed intervals of ticks. If the optical element had measurable displacement ideally at the nm level then the following equation would have given a measurable wavelength.

$$\lambda = \frac{2\Delta x}{N} - s$$

A point of error in this lab was human handling of moving the mirrors, rotating the ticks because

wavelengths operate on a very small scale and misaligning any optical axis or displacing optical lengths can drastically alter the fringe patterns. For precise accurate measurements, motorized control of mirrors would lead to quantitative observables.

4.2 Refractive Index Determination

Once again because of the limitations in the optical setup and a lack of a power meter to measure current the fringes were counted at fixed increments of degrees and subject to human error .

The Thorlab manual states that the laser light source is a HeNe laser which typically has a wavelength $\lambda \approx 632 \text{ nm}$ and applying a green filter makes $\lambda \approx 543 \text{ nm}$. By rotating the plate 6 degrees it was noted that the fringe pattern changed by 20. The plate has a different refractive index than the air medium so the beam path was refracted through the plexiglass. This explains the plate's effect on the interference pattern which obeys theory stating that any change in the refractive index in the beam arm paths affects the fringe pattern displayed on the screen. Shifting the optical length between the mirrors shifts the fringe by a corresponding wavelength.

$$\lambda = \frac{2\Delta x}{N}$$

Rearranging and noting $2\Delta x = \Delta s$

$$N\lambda = \Delta s$$

Therefore recording a 20 fringe difference roughly gives us 20λ path difference and in the case of a HeNe laser is roughly $12.64 \mu\text{m}$. However, this measurement is severely subject to human error, particles present in the optical set up, and even disturbances from human foot traffic. Minor changes in the refractive nature of the plate in the optical path are consistent with theory from the Thorlab's manual.

The refractive index of the 8mm plexiglass can quantitatively calculated. The equation provided

in the Thorlabs assumes the angle is fixed and the mirror is displaced to notice fringe pattern differences with a power meter.

$$n = \frac{\left(\frac{N\lambda}{2t} + \cos \alpha - 1\right)^2 + \sin^2 \alpha}{2\left(-\frac{N\lambda}{2t} - \cos \alpha + 1\right)}$$

Thor labs pg 31

Where $N = 20$, $t = 8\text{mm}$, and $\lambda = 543\text{ nm}$

This gives a refractive index of 1.14. Variations are subject to human error. However, this is consistent with theory in Thorlabs where the refractive index of the glass is higher than its surrounding air media of 1. This affects beam traveling through another media and affects the number of fringes present.

The optical path length for a path with a plate inserted is:

$$\Delta s = nt - t = t(n - 1)$$

$$\Delta s \text{ proportional to } t(n - 1)$$

From the data, it is calculated that the optical path is changed by $12.64\mu\text{m}$. Therefore from the relation the plexiglass refractive index is once again quantitatively above the refractive index of air.

4.3 Interference with LEDs

The laser has a longer coherence length because it maintains phase over a longer distance compared to the white and red LEDs.

$$L_c \approx \frac{\lambda_0^2}{\Delta\lambda} - 6$$

The change of lambda represents the spectral width and that of a laser is significantly lower

which gives a higher coherence length. In an optical set up this gives a forgiving range of values to view a visible pattern. Because the LEDs have higher spectral widths and give off light at varying wavelengths the coherence lengths are very short, demanding extremely precise set ups to capture the fringes. Because the path difference between the interferometer arms have to be within microns of one another for the LEDs, our group was unable to capture the fringes.

However, by corroborating with a group it was discovered that if we were able to precisely set up the optical alignment, the red LED's coherence length should have been slightly longer than the white LEDs coherent length. This would mean that red LEDs have a shorter spectral width compared to the white counterpart which would result in a slightly longer coherence length.

In addition, Thorlabs theory suggests that the red fringe will have stripes and dots appearing on the fringe pattern on the screen due to the scattering and diverging properties of the LED. Compared to the laser, the LED diverges light to the corners of the beamsplitter which produces unwanted stripes and dots on the screen's fringe pattern.

Because the white LED holds a larger spectrum of light from red to blue compared to its red LED counterpart. The longer wavelengths in colors such as red and shorter wavelengths in colors such as blue separate producing a rainbow pattern on the fringe pattern.

5 Conclusion

This experiment successfully demonstrated interference through fringe patterns through changing optical path lengths in the Michelson interferometer. Adjusting optical path lengths by displacing one mirror affected the number of fringes. Introducing heat in one of the arms in the interferometer changed the refraction in the optical path, producing oscillating fringe motion on the screen. Observing the fringe patterns on

two outputs of the interferometer produced contrasting destructive and constructive patterns based on the π phase difference. Rotating the plexiglas plate affected the fringe order, showing how a change in optical thickness can be observed through an interferometer. Although the LED fringes were unresolvable, it was consistent with theory of having small coherence lengths. To conclude, the experiment shows how varying optical length and changing coherence determines interference.

References

1. Saleh, B. E. A., & Teich, M. C. *Fundamentals of Photonics*, 2nd ed., Wiley, 2007.
2. Lecture Slides, “Week 4” Prof. Blumenthal, UCSB ECE 136A, Fall 2025.
3. Thorlabs, Inc. “Michelson Interferometer Educational Kit Manual.” Thorlabs, 2023. PDF.